

1. Hello.c

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
printf("Hello Linux\n");
```

```
return 0;
```

```
}
```

Compile: gcc hello.c -o hello

./hello

```
medikondarakesh@Medikondas-MacBook-Air ~ % nano Hello.c
medikondarakesh@Medikondas-MacBook-Air ~ % gcc hello.c -o hello
medikondarakesh@Medikondas-MacBook-Air ~ % ./hello
Hello Operating Systems!
medikondarakesh@Medikondas-MacBook-Air ~ % █
```

2. Read User Name

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
char name[30];
```

```
printf("Enter your name: ");
```

```
scanf("%s",name);
```

```
printf("Welcome %s\n",name);
```

```
return 0;
```

```
}
```

```
medikondarakesh@Medikondas-MacBook-Air ~ % gcc username.c -o username
medikondarakesh@Medikondas-MacBook-Air ~ % ./username
Enter your name: Rakesh
Welcome Rakesh
medikondarakesh@Medikondas-MacBook-Air ~ % █
```

3. Display Process ID

```
#include<stdio.h>
#include<unistd.h>
int main()
{
printf("Process ID = %d\n",getpid());
return 0;
}
```

```
medikondarakesh@Medikondas-MacBook-Air ~ % nano pid.c
medikondarakesh@Medikondas-MacBook-Air ~ % gcc pid.c -o pid
medikondarakesh@Medikondas-MacBook-Air ~ % █
```

```
4. Fork()
#include<stdio.h>
#include<unistd.h>
int main()
{
fork();
printf("Hello\n");
return 0;
}
```

```
medikondarakesh@Medikondas-MacBook-Air ~ % nano fork.c
medikondarakesh@Medikondas-MacBook-Air ~ % gcc fork.c -o fork
medikondarakesh@Medikondas-MacBook-Air ~ % ./fork
Hello
Hello
medikondarakesh@Medikondas-MacBook-Air ~ % █
```

```
5. Parent and Child
#include<stdio.h>
#include<unistd.h>
int main()
{
pid_t pid;
pid=fork();
if(pid==0)
{
printf("I am Child Process\n");
}
else
{
printf("I am Parent Process\n");
}
return 0;
}
```

```
medikondarakesh@Medikondas-MacBook-Air ~ % nano parentchild.c
medikondarakesh@Medikondas-MacBook-Air ~ % gcc parentchild.c -o parentchild
medikondarakesh@Medikondas-MacBook-Air ~ % ./parentchild
I am Parent Process
I am Child Process
medikondarakesh@Medikondas-MacBook-Air ~ % █
```

6. Display parent and Child PID

```
#include<stdio.h>
#include<unistd.h>
int main()
{
    pid_t pid;
    pid=fork();
    if(pid==0)
    {
        printf("Child PID = %d\n",getpid());
        printf("Parent PID = %d\n",getppid());
    }
    else
    {
        printf("Parent PID = %d\n",getpid());
    }
    return 0;
}
```

```
medikondarakesh@Medikondas-MacBook-Air ~ % nano piddisplay.c
medikondarakesh@Medikondas-MacBook-Air ~ % gcc piddisplay.c -o piddisplay
medikondarakesh@Medikondas-MacBook-Air ~ % ./piddisplay
Parent PID = 30327
Child PID = 30328
Parent PID = 30327
medikondarakesh@Medikondas-MacBook-Air ~ % █
```

7. Wait()

```
#include<stdio.h>
#include<unistd.h>
```

```
#include<sys/wait.h>
```

```
int main()
{
    pid_t pid;
    pid=fork();
    if(pid==0)
    {
        printf("Child Running\n");
    }
    else
    {
        wait(NULL);
        printf("Parent Running After Child\n");
    }
    return 0;
}
```

```
Parent PID = 88827
medikondarakesh@Medikondas-MacBook-Air ~ % nano wait.c
medikondarakesh@Medikondas-MacBook-Air ~ % gcc wait.c -o wait
medikondarakesh@Medikondas-MacBook-Air ~ % ./wait
Child Running
Parent Running After Child
medikondarakesh@Medikondas-MacBook-Air ~ % █
```

8. Wait()

```
#include<stdio.h>
```

```
#include<unistd.h>
```

```
int main()
```

```
{
```

```
printf("Before exec()\n");
```

```
execlp("ls","ls",NULL);
```

```
printf("After exec()\n");
```

```
return 0;
```

```
}
```

```
medikondarakesh@Medikondas-MacBook-Air ~ % nano exec.c
medikondarakesh@Medikondas-MacBook-Air ~ % gcc exec.c -o exec
medikondarakesh@Medikondas-MacBook-Air ~ % ./exec
Before exec()
2520030001_Practical      BufferedReader.java      For.java                  Numbers.java              Ruok.java
2520030001_Skill          Calculator.class          ForgeOS                   old.txt                   s.circ
2520030002_Practical      CardPayment.class        fork                      Os_rak                    script.sh
2520030002_Skill          Case.class               fork.c                    OS_cd                     ShoppingCart.class
2520030003_Practical      Case.java                hello                     OS_pwd                    shortcut
2520030003_Skill          Case.javacmd              hello.c                   parentchild               TestStratergy.java
2520030035_OSSP           data.txt                  hello.c                   parentchild.c             Time.class
2520030448_OSSP           ddcalab.circ             hello.c gcc hello.c -o hello paymentStrategy.class    Time.java
a.txt                     Desktop                  hello.java                Pictures                   TwoDimension.java
Applications              Dick].java               import java.util.java     pid                       Upipayment.class
Ar.java                   Documents                Library                    pid.c                     username
ark.txt                   Donor.class              logisim evolution         piddisplay                username.c
ArrayEx.java              Downloads                 Main.class                 piddisplay.c              wait
arrays.java               exec                      Main.java                  Public                      wait.c
b.txt                     exec.c                   Movies                      R.html                    WalletPayment.class
BloodBankSystem.class     exec.c:                  Music                      Rakesh
BloodBankSystem.java      file1.txt                my-app                     Red.c.save
Buffer.java               file2.txt                 npm                         report.txt
Buffered.java             For.class                 Numbers.class              Riot.txt
medikondarakesh@Medikondas-MacBook-Air ~ %
```

9)fork() + exec()

```
#include<stdio.h>
#include<unistd.h>
#include<sys/wait.h>
int main()
{
    pid_t pid;
    pid=fork();
    if(pid==0)
    {
        execlp("date","date",NULL);
    }
    else
    {
        wait(NULL);
        printf("Child Completed\n");
    }
    return 0;
}
```

```
medikondarakesh@Medikondas-MacBook-Air ~ % nano forkexec.c
medikondarakesh@Medikondas-MacBook-Air ~ % gcc forkexec.c -o forkexec
medikondarakesh@Medikondas-MacBook-Air ~ % ./forkexec
Sun Aug  9 11:31:41 IST 2026
Child Completed
medikondarakesh@Medikondas-MacBook-Air ~ % █
```